

Full paper

The influence of FEC on the solvation structure and reduction reaction of LiPF₆/EC electrolytes and its implication for solid electrolyte interphase formation

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ABSTRACT

Fluoroethylene carbonate (FEC) has been proposed as an effective electrolyte additive that enhances the stability and elasticity of the solid electrolyte interphase (SEI) of emerging Si and Li metal anodes. However, uncertainties still remain on the exact mechanism through which FEC alters the electrolyte decomposition and SEI formation process. Herein, the influence of FEC on LiPF₆/ethylene carbonate (EC) electrolytes for Si anodes is investigated through classical molecular dynamics, Fourier-transform infrared spectroscopy, and quantum chemical calculations. Albeit a minority species, FEC is found to significantly modify the solvation structure and reduction behavior of the electrolyte while being innocuous to transport properties. Even with limited 10% of FEC, the Li⁺ solvation structure exhibits a notably higher contact-ion pair ratio (14%) than the parent EC electrolyte (6%). Moreover, FEC itself, as a new fluorine-containing species, appears in 1/5 of the Li⁺ solvation shells. The Li⁺-coordinated FEC is found to reduce prior to EC and uncoordinated FEC which will passivate the anode surface at an early onset (ca. 0.3 V higher than EC) by forming LiF. The critical role of FEC in tailoring the Li⁺ solvation structure and as-formed protective SEI composition provides mechanistic insight that will aid in the rational design of novel electrolytes.

1. Introduction

To realize next-generation energy storage systems, there is a pressing need for functional, optimized electrolytes with excellent bulk stability and conductivity while exhibiting a suitable range of passivating reactions towards the chosen anode material [1,2]. Mounting evidence points to rational design of the bulk electrolyte solvation structure [3], including both majority as well as minority species, as a vehicle towards tailoring specific interfacial reactivity and reduction potentials of electrolyte components, which in turn contribute to the formation of a functional solid electrolyte interphase (SEI) layer.

An optimal SEI layer passivates the anode surface against further side reactions while facilitating Li-ion transport [4,5]. Extensive previous work shows that a complicated cascade of reduction reactions occurs during the first cycle, and that the initially formed SEI

containing inorganic species, e.g. LiF and Li₂CO₃ [6,7], as well as organic ones, e.g. lithium ethylene dicarbonates (LEDC) [8,9], further evolves through a variety of aging processes [10–13] (hydrolysis, reaction between Li salt and intermediate decomposition products, and thermal decomposition, etc.) as well as continuous electrolyte reduction [14,15]. The sequence of reactions and their aging depend both on the specific components and concentration, including additives and impurities, of the bulk electrolyte, and upon the specific anode material and its surface chemistry and structure. Hence, in principle, every anode material requires a differently tailored electrolyte, to achieve maximal compatibility—both electrochemically as well as mechanically.

One of the most promising anode materials for future high energy density Li-ion batteries (LIBs) is Si, due to its high theoretical specific capacity (exceeding 4200 mAh g⁻¹) and low cost [16]. However,

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conventional LIB electrolytes, such as lithium hexafluorophosphate (LiPF_6) in cyclic ethylene carbonate (EC) and linear dimethyl carbonate (DMC) and/or diethyl carbonate (DEC) [17,18,119], form a non-passivating Si SEI that is unable to mitigate the cracking due to Si's large volume expansion and contraction during cycling. To improve the Si SEI performance, several approaches [19] have been explored, e.g., nanostructuring [20,21], use of binders with tailored functionalities [22–24], as well as modified salts and electrolyte additives [25,26], fluoroethylene carbonate (FEC) has recently been proposed as an effective electrolyte additive that significantly enhances the stability and elasticity of the as-formed SEI film [27,28]. In the presence of FEC, which degrades at a higher reduction potential than both EC and DEC, a denser, more uniform and conformal SEI is formed on the silicon electrode [19]. This SEI layer has been found to ameliorate the emergence of large cracks and suppress further decomposition of EC/DMC, leading to enhanced electrochemical performance and improved coulombic efficiency [19]. To understand the composition and morphology of the Si SEI formed with FEC, extensive experiments using scanning transmission electron microscopy (STEM) [29], electron energy loss spectroscopy (EELS) [29], Fourier-transform infrared spectroscopy (FTIR) [25,30], X-ray photoelectron spectroscopy (XPS) [16,31], atomic force microscopy (AFM) [32,33], hard X-ray photoelectron spectroscopy (HAXPES) [19,34], nuclear magnetic resonance (NMR) [16,35], differential electrochemical mass spectrometry (DEMS) [36], and time-of-flight secondary ion mass spectrometry (ToF-SIMS) [37], etc., have been undertaken. Lucht [25] and co-workers recommended a 10% FEC inclusion based on a combination of low impedance and high capacity retention due to the formation of an SEI which contains both a flexible polymer and high lithium salt content (LiF and Li_2CO_3). Subsequent studies found that, compared with a standard EC/DMC electrolyte, the use of 10% FEC additive modifies the organic SEI components derived from LEDC and soluble linear oligomers to soluble and insoluble crosslinked poly(ethylene oxide)-based polymers (such as lithium poly(vinylene carbonate)), which could better passivate the anode surface and resist volume expansion [16,35,38]. Meanwhile, FEC incurs increased formation of LiF , less formation of Li_2CO_3 and organic carbonate species resulting in an overall lower interfacial impedance of the Si anode [39]. The presence of fluoride species further leads to the etching of the native oxide surface layer, improving the surface region Li conductivity and lowering the interfacial resistance [4,29]. Moreover, there is evidence that FEC influences the LiPF_6 decomposition reaction and may suppress further salt degradation after the initial cycles [19].

Numerous studies have endeavored to understand the reaction pathway and SEI formation mechanism through experiments [40] and multiscale computational simulations [41]. For example, experimental efforts have utilized Lithium Naphthalide (LiNap) as a one-electron reduction reagent [42,43] to analyze the resulting solid precipitates and gas evolution. Through this technique, FEC was found to decompose into a range of products including HCO_2Li , $\text{Li}_2\text{C}_2\text{O}_4$, Li_2CO_3 , and polymerized vinylene carbonate (VC), which supports a decomposition mechanism where FEC reduces to form VC and LiF , followed by subsequent VC reduction. Other complementary approaches include theoretical modeling the electrolyte [44], Si anode [45], and their interface [46,47]. Quantum chemical calculations confirm that defluorination reactions significantly increase the reduction potential of FEC [48]. Leung and Balbuena et al. have pioneered the *ab-initio* molecular dynamics (AIMD) simulation of EC and FEC on the surface of Si as well as lithiated Si (LiSi_4 , LiSi_2 , LiSi and $\text{Li}_{13}\text{Si}_4$) [49–52] including the native oxide layer [45,53], and proposed a series of possible decomposition mechanisms leading to LiF formation and polymerization. FEC is found to exhibit more diverse reaction pathways than the two-electron reduction of unsubstituted ethylene carbonate [54–56]. Both one- and two-electron reactions are feasible for the FEC reduction and result in a fluoride radical which in both cases contribute to the formation of LiF [52]. Also, it was found that radical species are

responsible for the electron transfer that allows SEI layer growth once its thickness has evolved beyond the electron tunneling regime [46,57,58]. While several studies have focused on the reduction mechanisms of single FEC [59] and simplified solvation/interface models [52,60], a thorough examination of the influence of FEC on the solvation structure of the electrolyte [61], and hence the reduction potentials of the associated majority as well as minority species whose populations are altered by FEC, has to our knowledge not been undertaken. For example, it has been widely assumed that FEC, exhibiting a lower donor number than EC, remains largely uncoordinated as an additive in LIB electrolyte formulations [62,63]. However, even as a minority species, FEC may strongly influence the reduction potential and decomposition mechanism of associated electrolyte components, and hence alter the onset as well as reaction sequence of the bulk electrolyte. One of the goals of this contribution is to investigate how FEC influences the solvation structure, which further determines the reduction reactions and the subsequent SEI formation process.

In the present work, we report classical molecular dynamics (MD) simulations coupled with first-principles quantum mechanics to describe the detailed solvation structure and reactivity of LiPF_6 in EC/FEC, as a function of FEC as well as salt concentration. The solvation structure, self-diffusion coefficient and other macroscopic properties of the EC/ LiPF_6 electrolyte with or without FEC additive are obtained, and the reduction potentials of the majority as well as minority species are calculated. Spin density analysis was used to further elucidate the reduction behavior of the solvate complexes obtained from the MD simulation, and hence the role of FEC in the SEI formation reaction. Corresponding experimental FTIR measurements are used for validation and as a direct determination of the coordination number in comparison with the calculated solvation structure. The aim of our work is to provide a deeper understanding of the subtle influence of the FEC additive on the bulk electrolyte and its constituents to aid in future rational design of functional electrolytes for Si anodes.

2. Calculation methods

2.1. Classical molecular dynamics (MD)

MD simulations were performed using the Large Scale Atomic/Molecular Massively Parallel Simulator (LAMMPS) [64] code for electrolytes of 1.0/1.2 M LiPF_6 in EC or EC with 5/10 mol% FEC. Although cyclic solvents are often used together with linear solvent ingredients such as DEC or DMC to modify viscosity, it was found that when EC contributes more than 50% by weight of the electrolyte, EC becomes, to good approximation, the exclusive “inner member” of the Li^+ solvation sheath [65]. Hence, only EC (and FEC) molecules are considered here to simplify the model. 1500 solvent molecules were used as bulk electrolytes. The number of salt molecules was adjusted via several test runs for 1 atm and 298 K equilibrium conditions to achieve the 1.0 M and 1.2 M concentrations. Specifically, the final number of salt molecules were 104 and 126 with equilibrated box volume of 177.5 nm^3 and 180.2 nm^3 , respectively. The molecules were initially packed randomly in a cubic box of size $54 \times 54 \times 54 \text{ \AA}^3$ periodic in the XYZ direction using PACKMOL [66] (Figure S1, available in the supplementary material). The initial configuration was minimized by conjugated-gradient energy minimization scheme employing a convergence criterion of 1.0×10^{-4} . While the thermodynamic melting point for EC with 1 M LiPF_6 is approximately 298 K [67], the kinetic liquid range limit may be lower [68], which allows a room temperature liquid state simulation. The systems were equilibrated for 2 ns in the isothermal-isobaric ensemble (constant NPT) using the Parrinello–Rahman barostat to maintain a pressure of 1 bar and a temperature of 298 K with a time constant of 1 ps. An annealing process was conducted to further guarantee that all systems are melted and to avoid local configuration confinement. All systems were heated from 298 K to 400 K for 1 ns, and maintained at 400 K for 1 ns, and subsequently annealed from 400 to 298 K in 1 ns.

Finally, the production runs of 5 ns were conducted in the canonical ensemble (NVT) under Nose-Hoover thermostats with a time constant of 1 ps at 298 K. The simulation time was long enough to sample adequately the Fickian (diffusive) regime of all systems, which was justified by a 55 ns long run (Figure S2).

A non-polarizable force field model was employed, which is defined by the following potential functions:

$$E_{total}(r^N) = E_{bonds} + E_{angles} + E_{dihedrals} + E_{impropers} + E_{nonbonded} \quad (1)$$

$$E_{bonds} = \sum_{bonds} K_r(r - r_0)^2 \quad (2)$$

$$E_{angles} = \sum_{angles} K_\theta(\theta - \theta_0)^2 \quad (3)$$

$$E_{dihedrals} = \sum_{dihedrals} V [1 + \cos(n\phi - d)] \quad (4)$$

$$E_{impropers} = \sum_{impropers} V [1 + d\cos(n\phi)] \quad (5)$$

$$E_{nonbonded} = \sum_{i>j} 4\epsilon_{ij} \left[\left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}} \right)^6 \right] + \sum_{i>j} \frac{Cq_i q_j}{\epsilon r_{ij}}, \quad r < r_c,$$

$$\epsilon_{ij} = \sqrt{\epsilon_{ii}\epsilon_{jj}}, \quad \sigma_{ij} = \frac{\sigma_{ii} + \sigma_{jj}}{2} \quad (6)$$

The bonded interactions (bonds, angles, dihedrals, and impropers) were modeled as harmonic functions and the nonbonded included van der Waals interactions and Coulombic forces. The bonded and nonbonded parameters for EC and FEC were obtained from the Optimized Potentials for Liquid Simulations All Atom (OPLS-AA) force fields [69,70], the PF_6^- anion from Lopes et al. [71], and the lithium cations from Jensen et al. [72]. The partial atomic charges for all molecules were derived by first optimizing the geometry using Becke's three-parameter exchange function combined with the Lee-Yang-Parr correlation functional (B3LYP) [73] at the aug-cc-pvdz theory level using the Gaussian 16 [74] package and then fitting the electrostatic potential surface using the RESP method [75,76]. Long-range electrostatic interactions were handled by the particle-particle particle-mesh (PPPM) solver with a grid spacing of 0.1 nm. A cutoff distance of 1.5 nm was used for electrostatic and 12–6 Lennard-Jones interactions. Moreover, each Li ion in the system is surrounded by at least one PF_6^- in the first solvation shell or second solvation shell according to the trajectories (Figure S3). Hence, if we ignore the negligible amount of aggregates (AGG), all Li ions can be categorized into the contact ion pair (CIP) and solvation separated ion pair (SSIP). A Boltzmann factor was used to estimate the CIP formation free energy from the population difference of the CIP and SSIP structures.

$$\Delta_f G_{CIP} = -k_B T \ln \frac{p(CIP)}{p(SSIP)} \quad (7)$$

where p is the population for each species, $\Delta_f G_{CIP}$ is their relative free energy, k_B is the Boltzmann constant and T is the temperature. For NVT run under equilibrated pressure, the Helmholtz free energy (A) is approximately equal to the Gibbs free energy (G).

To further examine the influence of FEC on the bulk electrolyte, transport properties are characterized by the self-diffusion constant (D) and the transference number (t_{Li^+}). The self-diffusion coefficients for Li^+ , PF_6^- , EC, and FEC were extracted from the MD simulation by analysis of the mean square displacement (MSD) over time using the Stokes–Einstein relation [77]. The slope of the linear regime in the MSD was obtained for each simulation duration of 1 ns and averaged over 5 ns of the production runs to obtain:

$$D = \frac{1}{6} \frac{\langle (\delta r)^2 \rangle}{\Delta t} \quad (8)$$

Based on the self-diffusion constants, the Li^+ ionic transference

number t_{Li^+} , was calculated from ratios of D according to Ref. [78]:

$$t_{Li^+} = \frac{D_{Li^+}}{D_{Li^+} + D_{PF_6^-}} \quad (9)$$

2.2. Quantum chemical calculations

Geometries of solvate complex were optimized from the initial structures observed in MD simulations to obtain their theoretical IR vibrations using Gaussian 16 at the B3LYP/6–311++G(d) level of theory [79]. The calculated IR data was then accumulated according to the composition ratio from the MD simulations to obtain the total spectra of each electrolyte. The adiabatic reduction potentials for the representative solvation structures were calculated using the following function [80].

$$E_{adiabatic} = -(G_{reduced} - G_{initial} + \Delta G_{solv}^o(reduced) - \Delta G_{solv}^o(initial)) / F - 1.4V \quad (10)$$

where $G_{reduced}$ and $G_{initial}$ are the free energies of the reduced and initial complexes at 298.15 K in gas-phase at 298.15 K, respectively; ΔG_{solv}^o are the corresponding free energies of solvation, and F is the Faraday constant. The zero-point energy (ZPE) corrections were considered in the calculation while the basis set superposition error (BSSE) energy was neglected [81]. The solvent effects in the free energy calculation of each complex were considered using both an explicit cluster model and an implicit polarizable continuum model (PCM) [82]. The inclusion of empirical dispersion correction [83] resulted in very slight changes in reduction potentials ≤ 0.05 V (Table S1), and therefore was not included. A standard-state correction [84] was also considered to account for the four reduction reactions with free EC or PF_6^- generation, which resulted in the addition of a correction constant ($R \ln(24.47/M)$), where R is the gas constant, T is temperature, 24.47 is the molar volume in liter for ideal gas under 1 atm and 298.15 K, $M = 1$ for PF_6^- , $M = 0.07$ for EC) to ΔG_{solv}^o . A dielectric constant of 90 was adopted for the EC solvent as well as the EC/FEC mixture [85]. Geometries were allowed to relax after the electron transfer. Subtraction of 1.4 V accounts for the conversion from the absolute electrochemical scale to the commonly used Li/Li^+ potential scale in order to compare predicted values with experimental data using the same reference electrode. An additional factor of ca. 0.1 V for graphite intercalation or 0.3–0.4 V for Si anode Li insertion should be subtracted if the reference electrode is changed to these specific systems. The spin density calculation of the reduced state structures was conducted using natural bond orbital (NBO) theory. The dipole moment calculation was conducted at the B3LYP/6–311++G(d,p) level of theory.

3. Experimental methods

3.1. Fourier-transform infrared spectroscopy (FTIR)

The FTIR spectra for 1.0 M LiPF_6 in EC and 1.0 M LiPF_6 in EC with 10 wt% FEC were collected with a FTIR spectrometer (Bruker, ALPHA) using a diamond attenuated total reflection (ATR) accessory. Two pure solutions (EC and EC w/10 wt% FEC) were also tested as a comparison. Spectra were collected in the region from 4000 to 650 cm^{-1} with 128 scans and 2 cm^{-1} resolution in an argon-filled glovebox with O_2 and $\text{H}_2\text{O} < 0.1$ ppm. Ethylene carbonate (EC, anhydrous, $\geq 99\%$), lithium hexafluorophosphate (LiPF_6 , battery grade) and fluoroethylene carbonate (FEC, 99%) were purchased from BASF. All spectra were normalized via stretching band of $-\text{CH}_2$ at 3000 cm^{-1} . This analysis is based upon the assumption that the infrared intensities of the uncoordinated and coordinated structures are equivalent after normalization (no scaling of the bands).

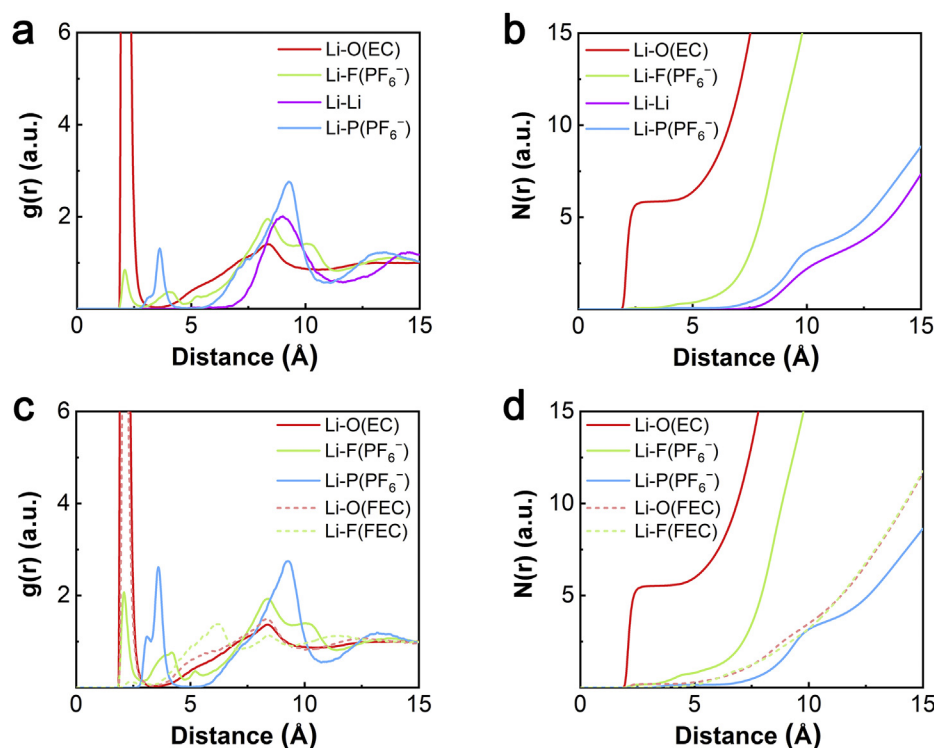


Fig. 1. Calculated radial distribution functions, $g(r)$, and the corresponding integrals, $N(r)$, of Li-O(EC), Li-F(PF₆⁻), Li-Li, Li-P(PF₆⁻) pairs of (a) (b) 1.0 M LiPF₆ in EC, and Li-O(EC), Li-F(PF₆⁻), Li-P(PF₆⁻), Li-O(FEC), Li-F(FEC) pairs of (c) (d) 1.0 M LiPF₆ in EC with 10%mol FEC additive.

4. Results and discussion

4.1. Solvation structure

A sequential simulation for 1.0/1.2 M LiPF₆ in EC with 0/5/10% FEC was conducted to investigate the influence of FEC on the solvation structure of the LiPF₆/EC electrolyte. The population proportion of solvent molecule and anion in the first solvation shell of Li⁺ is clearly altered after adding FEC, which is confirmed by the radial distribution function, $g(r)$, and the corresponding integrals, $N(r)$, of Li-X (X = O (EC), F(PF₆⁻), Li, P(PF₆⁻) for EC and X = O(EC), F(PF₆⁻), O(FEC), P(PF₆⁻), F(FEC) for EC w/FEC) pairs as shown in Fig. 1, Figure S4, and Table S1. The dominant peak of the Li-F pair at ca. 8 Å shown in Fig. 1a suggests that for 1.0 M LiPF₆ in EC, most of the LiPF₆ salt forms solvent separated ion pairs (SSIP), while the contact ion pairs (CIP) present as minority species (Fig. 2b). In addition, a trace amount (less than 1%) of aggregate solvates (AGG) were also observed during the simulation. By integrating the $g(r)$ to 3.0 Å, the total coordination number (CN) for Li ion is obtained as shown in Fig. 2a. Interestingly, while most of the first solvation shell is occupied by EC solvent molecules (5.84 out of 5.90) with a most probable distance of 2.08 Å to Li⁺, the Li⁺ solvation sheaths also present an average of 0.06 PF₆⁻ anions, most of which form monodentate structures coordinated through a fluorine. The simulation produced a longer distance of 2.11 Å for the Li-F pair, illustrating that EC molecules take closer positions in solute complexes. The AGG species are rare, indicating an almost negligible number of Li ions that coordinate with more than one PF₆⁻ anion. Hence, the corresponding CIP ratio is estimated as 6%, with a SSIP ratio of 94%, which agrees qualitatively with previous MD simulations [86], and is in agreement with common assumptions that LiPF₆ is a weakly coordinating salt compared to LiBF₄ and well dissociated in EC solutions [87].

Upon addition of 10% FEC into the LiPF₆/EC electrolyte, the solvation structure is evidently altered in several ways, primarily due to the weakened donor ability of FEC. Besides the similar solvation

structures as shown in Fig. 2b, other solute complex structures containing FEC were also observed in the snapshots of the MD simulation (Fig. 2c). Intriguingly, it is the carbonyl O in FEC that binds with the Li ion rather than F, which means that the composition difference of the solvent molecule does not radically alter the binding behavior as compared with the parent EC molecule [63]. However, the average CN for Li-O(EC) pair decreases from 5.84 to 5.51, allowing for an average Li-O(FEC) pair contribution of 0.19 to the first solvation shell. A maximum in $g(r)$ of 2.12 Å was observed for the Li-O(FEC) pair, suggesting a weaker interaction between Li-FEC as compared to Li-EC. Notably, the CIP ratio doubles from 6% to 14%, which significantly changes the statistics of the SEI formation reaction precursors, indicating a prominent role of FEC, even as a minority species. Similar to the EC electrolyte, the simulation with FEC additive results in a same Li-EC distance. Yet, the Li-FEC distance is slightly farther away (2.12 Å) as compared to the Li-EC counterparts, which results in a ‘looser’ structuring of the first solvation shell. EC electrolytes are known to promote salt dissociation due to its high dipole moment (5.64 Debye, calculated), as well as a high donor number. In contrast, FEC exhibits a lower dipole moment (4.97 Debye, calculated) which results in less dissociation of LiPF₆ and weakened donor ability (i.e. smaller CN of Li⁺-solvent). This was further illustrated by inspecting the different charge population on the carbonyl oxygen of each molecule. Natural bond orbital analysis results in a charge on O_{EC} of -0.66, while the charge on O_{FEC} is -0.62, indicating the weaker donor ability of FEC as compared to EC. If we compare all six simulated compositions listed in Fig. 2a, it is obvious that a higher LiPF₆ concentration gives a higher CIP ratio. More importantly, when FEC is added to the system, a non-negligible percentage of the solute complexes are modified with fluorine-containing FEC that occupy the first solvation shell. The tendency for CIP formation can be represented by the free energy ΔG_{CIP} which is obtained through the CIP population, assuming a Boltzmann distribution of electrolyte species. Table 1 summarizes the results which show that higher salt concentration as well as inclusion of FEC reduces the energy cost of contact ion pairing. Previous first-principles calculations

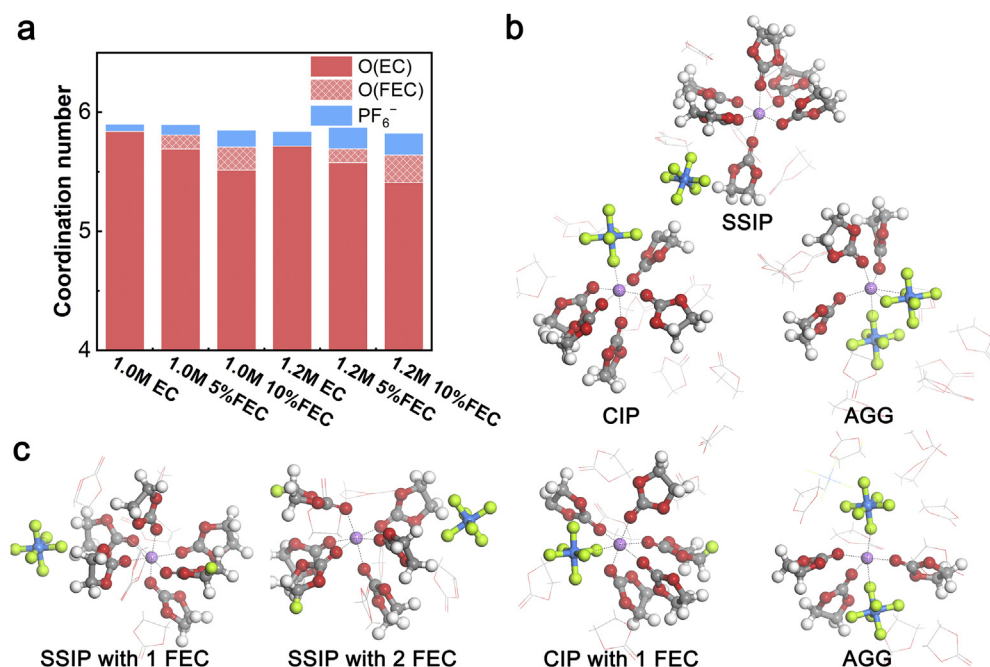


Fig. 2. (a) The calculated total coordination number for Li^+ in 1.0 M LiPF_6 in EC with 0/5/10% FEC and 1.2 M LiPF_6 in EC with 0/5/10% FEC with specifying the contributions from EC, FEC and PF_6^- . The representative solvation structures taken from MD simulation snapshots of (b) 1.0 M LiPF_6 in EC and (c) 1.0 M LiPF_6 in EC with 10% FEC. The carbon, hydrogen, oxygen, fluorine, phosphorus, and lithium elements are represented by grey, white, red, green, blue, and purple, respectively.

reported a $\Delta_f G_{\text{CIP}}$ of $2.6 \text{ kcal mol}^{-1}$ for the $\text{Li}^+ - (\text{PC})_4$ to $\text{Li}^+ - \text{PF}_6^- (\text{PC})_3$ conversion [88], which is approximately 1 kcal mol^{-1} higher than that of 1.0 M LiPF_6/EC obtained here from the CIP population. We speculate that the inclusion of explicit solvation effects beyond the first solvation shell as well as polarization effects not included in the current simulations may contribute to the difference between the two methodologies. In particular, the inclusion of 10% of FEC further reduces the solvating strength of the composite electrolyte, and allow for more CIPs, i.e. for PF_6^- to directly contact/interact with Li^+ . The coordination number for the Li-FEC pair increases to 0.2, which means that about 20% of the solvate structures on average now contain at least one FEC. We argue that this content change of the solute complex significantly changes the reduction potentials of the electrolyte constituents, which will be addressed further in the following sections. We also note that, with a higher temperature, the total coordination number of Li^+ is lower [89]. Therefore, a test of the electrolyte system under a sequential temperature of 330 K, 350 K, 400 K were performed to validate the results (Table S2). It was found that, with higher temperature, the CN for $\text{Li}^+ - \text{EC}$ decreases from 5.84 (298 K) to 5.23 (400 K), and the population of $\text{Li}^+ - \text{PF}_6^-$ CIPs increases from 6% (298 K) to 25% (400 K). This may be explained by the increase in random thermal motion of solvent molecules resulting in a decrease of the dielectric constant and weakened EC-Li (dipole-charge) interactions. Thus, the Li- PF_6^- (charge-charge) interaction would be gradually favored with elevated T.

4.2. Transport properties

Based on the mean square displacement (MSD) of each component during the simulation (Figure S5), the self-diffusion coefficients and transference numbers were calculated and plotted along with the reference experimental results [90] in Fig. 3. Both 1.0 M and 1.2 M simulation results indicate that the diffusion coefficients of all components within the EC electrolyte and EC/FEC mixture exhibit similar values (Fig. 3a). The trends for the three components EC, PF_6^- and Li^+ are in good agreement with the simulation results [89] obtained by a generalized AMBER force field (GAFF) [91] as well as the trend from NMR results by Hayamizu et al. [90] such that solvent diffusivities are $2\text{--}5 \times$ larger than ion diffusivities. While all the simulations predict slower computed dynamics than the experimental results, the difference is within one order of magnitude ($10^{-10}\text{--}10^{-11} \text{ m}^2 \text{ s}^{-1}$), illustrating that the classical MD simulation is adequately accurate to reproduce or predict the dynamical property trends of this system. It is noteworthy that, when FEC is added into the system, there is no significant variation of transport properties. EC and FEC exhibit almost the same diffusion coefficients within the numerical uncertainties and we surmise that the intermolecular interaction behaviors of EC and FEC are similar due to the common cyclic carbonyl structures with only one substituent. Hence, adding FEC does not directly affect the transport property of the electrolyte. On the other hand, FEC inclusion implicitly promotes the formation of CIPs while still remaining a minority species in the electrolyte. According to the Nernst-Einstein relation [92], an 8% CIP ratio increase from EC electrolyte to EC with 10% FEC will lead to

Table 1

The calculated coordination number of Li-X pairs, total coordination number, CIP ratio and corresponding contact ion pair (CIP) formation free energy $\Delta_f G_{\text{CIP}}$ of 1.0 M LiPF_6 in EC with 0/5/10% FEC and 1.2 M LiPF_6 in EC with 0/5/10% FEC.

Electrolyte	Coordination number					CIP ratio	$\Delta_f G_{\text{CIP}}$ (kcal mol $^{-1}$)
	O(EC)	O(FEC)	P(PF_6^-)	F(PF_6^-)	Total		
1.0 M EC	5.84	–	0.06	0.07	5.90	6%	1.63
1.0 M EC w/5%FEC	5.69	0.12	0.09	0.09	5.90	9%	1.37
1.0 M EC w/10%FEC	5.51	0.19	0.14	0.17	5.85	14%	1.07
1.2 M EC	5.71	–	0.12	0.20	5.84	12%	1.18
1.2 M EC w/5%FEC	5.58	0.12	0.18	0.20	5.87	18%	0.90
1.2 M EC w/10%FEC	5.41	0.23	0.18	0.23	5.82	18%	0.90

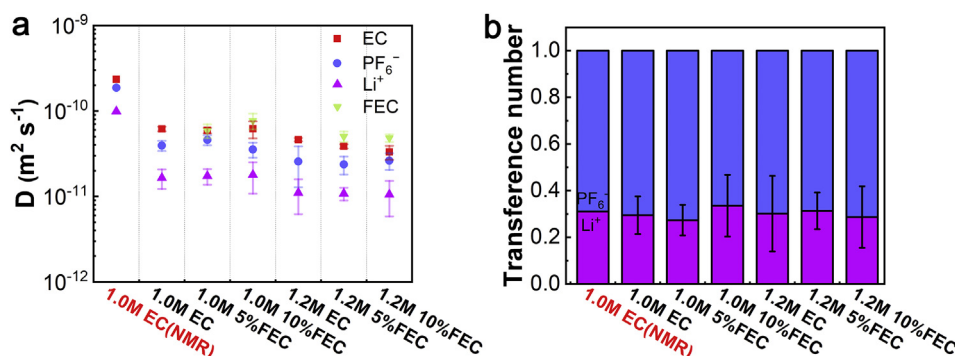


Fig. 3. (a) Self-diffusion coefficients computed from MD simulations at 298 K as compared with NMR experiments (1.0 M LiPF₆ in EC) from Hayamizu et al. [90] (b) Transference numbers for Li⁺ and PF₆⁻ from MD simulations and NMR experiments from Hayamizu et al. [90] The error bars represent the standard deviation of the data collected every 1 ns during the 5 ns production runs.

approximately the same percentage (8%) decrease in Li⁺ ionic conductivity. In contrast, the salt concentration clearly affects the ionic diffusivity, decreasing by an average of ca. 30% from 1.0 to 1.2 M, in agreement with previous studies [86]. As for the transference number, all the six electrolyte systems give similar results and coincide well with the NMR experimental value of 0.35 [90]. Hence, both the self-diffusion coefficient and the transference number results demonstrate a weak dependence of transport properties on the FEC additive up to 10%, elucidating that FEC modifies the solvation structure without greatly affecting the ion transport ability.

4.3. FTIR measurements

In order to verify the theoretical simulation results, we conducted FTIR measurements for a range of EC and EC/FEC electrolytes [93]. The peaks from the FTIR spectrum are deliberately deconvoluted to quantitatively estimate the population proportion of each electrolyte species. In parallel, we performed quantum chemical calculations to obtain the theoretical IR vibrational information for each solvation structure observed in the MD simulations. The calculated spectral profiles are further linearly combined using the population ratios obtained from MD simulations which result in predicted theoretical IR spectra. In general, there are four major characteristic peaks [94] that are shifted after the inclusion of FEC (Figure S6): (1) the C=O stretching band at 1760–1800 cm⁻¹, (2) the C=O breathing band at 710–730 cm⁻¹, (3) the P–F stretching band at 840–880 cm⁻¹, as well as (4) the ring breathing band at 890–910 cm⁻¹. The C=O breathing vibrational band at 710–730 cm⁻¹ can be used to distinguish the responses from different C=O binding states. Experimental peaks (Fig. 4a and 4b) were designated as coordinated and uncoordinated based on the calculations as well as reference data [94], which provide detailed information of the solvation structure. First, we note that the experimental results are in good agreement with the calculated spectra (Fig. 4c and d, and S7), and identical peaks are found in the spectra from both methods, despite small deviations in the absolute frequency values. For the EC electrolyte (Fig. 4a), two peaks at 728 cm⁻¹ and 715 cm⁻¹ were identified as coordinated EC and uncoordinated EC, both accompanied by a small C–H deformation peak at 706 cm⁻¹ [95]. By analyzing the measured peak area integrals, it was found that 38.1% of EC molecules are coordinated with Li⁺, which corresponds to a CN of 5.13 for Li–EC. While the calculated CN (5.84) from MD simulations is slightly higher than the experimental result, notably, both approaches indicate a CN over 5 for the EC electrolyte system. When 10% of FEC is included, an additional peak arises at 738 cm⁻¹, corresponding to the coordinated FEC. Meanwhile, a free FEC breathing band at 729 cm⁻¹ overlaps with the coordinated EC vibration band (Fig. 4b), which makes it challenging to deconvolute the peaks and obtain the exact percentage of the coordinated EC and uncoordinated FEC. By comparing the peaks of the uncoordinated EC, we estimate an increase in uncoordinated EC area by 6.6% as compared to the 1 M LiPF₆ in EC system, further supporting the decrease in CN of Li⁺–EC with respect to the EC electrolyte. Additionally, the CN of FEC

to Li⁺ is directly calculated as 0.21 from the green area, which corresponds well with our previous calculation results (0.19) from the MD simulations.

In addition to the CN of solvents, the tendency for contact ion pairing was probed by analyzing the P–F bond stretching band. As shown in Fig. 5, the peak at 838–840 cm⁻¹ is identified as the response from uncoordinated PF₆⁻, while the two peaks at 878 cm⁻¹ and 834 cm⁻¹ arise through Li⁺-coordinated PF₆⁻. In the EC electrolyte (Fig. 5a), coordinated PF₆⁻ contributes 6.4%, showing an excellent agreement with our simulated 6% CIP ratio. With the inclusion of FEC in the EC electrolyte (Fig. 5b), an FEC ring deformation vibration band appears at 862 cm⁻¹, which complicates the integral area calculation. By comparing the area change of the uncoordinated peak (Figure S8), we estimate that approximately 15% of the total PF₆⁻ ions coordinate with Li⁺ after adding FEC. In summary, both experimental and simulation results suggest that the CIP ratio, ca. 15% and 14% respectively, is increased under the influence of FEC additive.

The C=O bond stretching at 1790–1810 cm⁻¹ is considered characteristic of the binding state of the carbonyl group [94]. However, an overtone peak of the ring breathing band appears at the same position in the same area (Figure S9) [95–97]. While the C=O stretching vibration is here calculated to be a single band at 1808 cm⁻¹ by first-principles, experimentally, it overlaps with the Fermi resonance of an overtone of the ring in-plane breathing band (893 cm⁻¹). When solvent molecules coordinate to Li⁺, both these peaks are shifted, preventing further meaningful interpretation. As for the ring breathing band at 893 cm⁻¹, a sharp, blue-shifted signal at 904 cm⁻¹ appears upon coordination (Figure S10). The intensity of the shifted ring breathing band is significantly increased as compared to the original peak and hence prevents qualitative analysis. Consequently, due to these complexities, the C=O bond stretching and the ring breathing bands are disqualified for quantitative analysis of the solvation structure. In summary, the FTIR experiments and the MD simulations provide consistent solvation structure information. Both approaches demonstrate that the LiPF₆/EC electrolyte with or without FEC results in a Li⁺ CN of 5–6. We note that while carbonate-based electrolytes are traditionally believed to exhibit majority species corresponding to tetrahedrally coordinated carbonyl oxygen atoms around Li⁺ [89,98,99], the coordination number and solvation structure in these solvents systems are still under debate [99]. Recent results have reported a CN of 5–6 from two-dimensional infrared spectroscopy [100], ¹³C NMR [88,101], ¹⁷O NMR [102], and MD simulation [86], in agreement with the results in this contribution. In contrast, several previous AIMD simulations [103–105] suggest tetrahedral Li⁺-solvent coordination in the first coordination shell. However, we note that the typical simulation time in AIMD simulation (< 1 ns) is less than the residence time of a typical lithium-anion/solvent pair [106,107] and hence it may not capture the equilibrium state. In addition, Borodin et al. [106] also reported a coordination number of 4 for 1 M LiPF₆/EC using a polarizable force fields model. However, a very high contact ion pair ratio of 80% was also observed in the simulation, contrasting the perception of LiPF₆ as a weakly coordinating

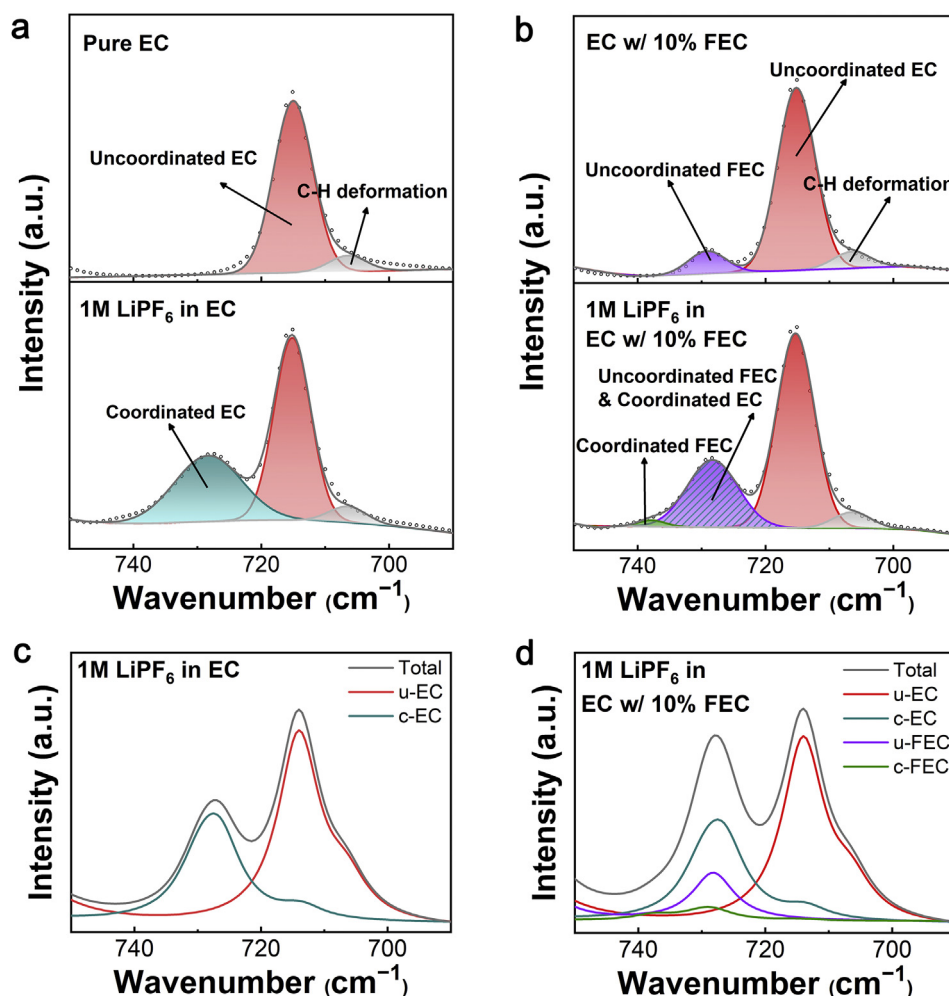


Fig. 4. Measured FTIR spectra of the C=O breathing band of (a) pure EC and 1.0 M LiPF₆ in EC, and (b) EC with 10% FEC and 1.0 M LiPF₆ in EC with 10% FEC. (c–d) The corresponding calculated IR spectra in comparison with the experimental results. Red, cyan, purple, green, grey and dark grey lines correspond to uncoordinated EC, EC coordinated with Li⁺, uncoordinated FEC, FEC coordinated with Li⁺, C–H deformation, and total spectrum, respectively. Scatter points denote the original FTIR data points.

salt [87].

Furthermore, our results conclusively suggest that adding FEC increases the CIP ratio while also adding another fluorine-containing species into the first solvation shell, without significantly impacting the transport properties of either Li⁺ and PF₆[−] ions. This minority FEC coordination with Li⁺ is important, as the Li⁺ solvate complex serves as a key precursor for electrode surface reduction reactions. The inclusion of FEC in the first solvation shell, even as a minority species, increases the reduction potential of FEC, due to its close proximity of Li⁺, as compared to a freely solvated FEC. Hence, to further investigate the critical influence of FEC on the SEI formation process, through the solvation structure of the electrolyte, we performed first-principles calculations of the reduction potential of the solvate structures obtained from the MD simulations and analyzed their preliminary reduction products.

4.4. Reduction potentials

The reduction potentials of free EC and FEC, their corresponding Li⁺ (solvent) complexes, and Li⁺–PF₆[−] (solvent) complexes were investigated at the B3LYP/6–31 + g(d) level of theory (Table 2). To effectively compare with experimental results, the calculated potential values were convoluted with an arbitrary 0.1 V width concave triangular wave, and plotted together with the experimental differential capacity (dQ/dV) versus potential (V) profile (Fig. 6) [108]. The

reduction products after geometry optimization were further scrutinized through spin density analysis to elucidate the reduction reaction process (Fig. 7). The free EC molecule exhibits a calculated reduction potential of 0.21 V, with most of the extra electron residing on the O=CO(O) moiety which consequently deforms out of plane. According to Natural Bond Orbital (NBO) analysis, the C atom hybridization state changes from sp^{2.0} to sp^{2.6} to accommodate the extra e[−]. The coordinated EC exhibits an increased reduction potential of 0.44 V–0.59 V (within 0.15 V), as compared with the uncoordinated molecule, which contributes to a broadening of the reduction peaks in the differential capacity plot. The obtained reduction potential is in good agreement with the previously obtained value of 0.53 V by G4MP2 [59]. Furthermore, the electron density of the reduced coordinated EC resides in the same region (e.g., on the O=CO(O)) as compared to the uncoordinated reduced EC. (Fig. 7c, h, i, k, and l). We find weak to little dependence of the reduction potential on the number of explicitly coordinating solvents [59,109], e.g., the calculated corresponding reduction potentials are 0.50 V and 0.59 V for 5- and 6- fold structures (Fig. 7k, and l), respectively. However, when Li⁺ is in direct coordination with FEC, the predicted FEC reduction potential is obtained as 0.81 V–0.91 V, which is about 0.3 V higher than uncoordinated FEC. The calculated reduction potentials are in good agreement with the previous G4MP2 calculations (0.90 V) and measured values (1.0 V) [110]. If we compare the reduction behavior of EC-containing species and FEC-containing species, even the uncoordinated FEC exhibits an

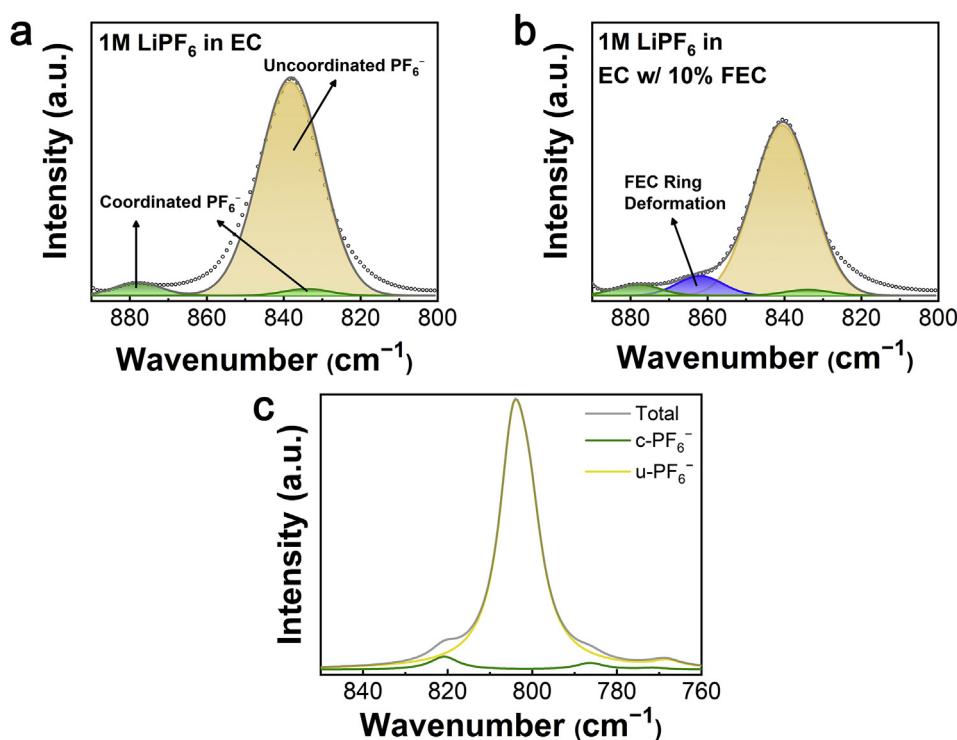


Fig. 5. Measured FTIR spectra of the P-F bond stretching band of (a) 1.0 M LiPF₆ in EC, and (b) 1.0 M LiPF₆ in EC with 10% FEC. (c) The corresponding calculated IR spectra for 1.0 M LiPF₆ in EC in comparison with the experimental results. Yellow, green, blue, and dark grey lines correspond to uncoordinated PF₆⁻, coordinated PF₆⁻, FEC ring deformation, and total spectrum, respectively. Scatter points denote the original FTIR data points.

Table 2

The reduction potential vs Li⁺/Li(s) (i.e. subtract 1.4 V) of individual solvent molecules and solvate complexes, in Volt, where *corr.* denotes values after the aforementioned standard-state correction.

Structures	Reduction potential
EC + e ⁻ → EC ⁻	0.21 V
FEC + e ⁻ → FEC ⁻	0.59 V
Li ⁺ - EC + e ⁻ → Li ⁺ - (EC) ⁻	0.54 V
Li ⁺ - FEC + e ⁻ → Li ⁺ - (FEC) ⁻	0.90 V
Li ⁺ - (EC) ₄ + e ⁻ → Li ⁺ - (EC) ₃ (EC) ⁻	0.49 V
Li ⁺ - (EC) ₃ (FEC) + e ⁻ → Li ⁺ - (EC) ₂ (FEC)(EC) ⁻	0.55 V
Li ⁺ - (EC) ₃ (FEC) + e ⁻ → Li ⁺ - (EC) ₃ (FEC) ⁻	0.91 V
Li ⁺ - (EC) ₅ + e ⁻ → Li ⁺ - (EC) ₃ (EC) ⁻ + EC	0.50 V (corr.)
Li ⁺ - (EC) ₆ + e ⁻ → Li ⁺ - (EC) ₃ (EC) ⁻ + 2EC	0.59 V (corr.)
Li ⁺ - (EC) ₅ (FEC) + e ⁻ → Li ⁺ - (EC) ₃ (FEC) ⁻ + 2EC	0.81 V (corr.)
Li ⁺ - PF ₆ ⁻ + e ⁻ → Li ⁺ - F ⁻ + PF ₅	spontaneous bond breaking
Li ⁺ - PF ₆ ⁻ (EC) + e ⁻ → Li ⁺ - PF ₆ ⁻ (EC) ⁻	0.59 V
Li ⁺ - PF ₆ ⁻ (FEC) + e ⁻ → Li ⁺ - PF ₆ ⁻ (FEC) ⁻	0.90 V
Li ⁺ - PF ₆ ⁻ (EC) ₅ + e ⁻ → Li ⁺ - (EC) ₃ (EC) ⁻ + PF ₆ ⁻ + EC	0.44 V (corr.)

equal or higher reduction potential than all investigated EC species, indicating a preferential reduction for FEC vs EC—for both minority as well as majority species. While the optimized FEC reduction products (Fig. 7b, d, j, and m) exhibit the same electron distributions as its EC counterparts, the stronger exothermic nature of the FEC reduction indicates that these FEC-derivative species are sufficiently metastable to decay through reactions other than the C-O bond ring-opening of EC [27]. This major mechanistic difference between the non-fluorinated and fluorinated carbonates allows for the recombination of fragments and intramolecular electron migration, facilitating the subsequent polymerization and LiF formation [27]. Previous DFT calculations have shown that the additional F fragments from FEC decomposition exhibit a “glue effect” by strongly binding to Li atoms of multiple organic species and connecting them [63], leading to a more compact and stable SEI. The formation of Li⁺-coordinated FEC species is especially important because in their absence, there is no early onset FEC reduction. Indeed, uncoordinated FEC is predicted to reduce at a similar

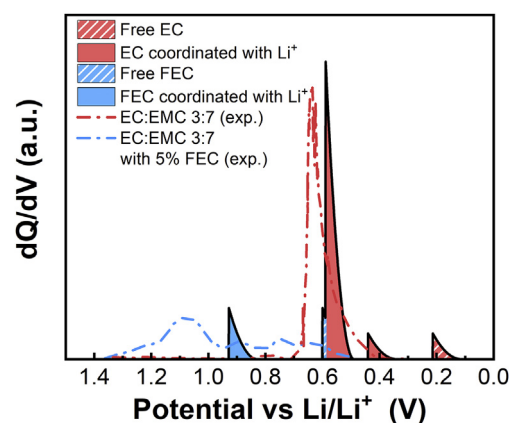


Fig. 6. The calculated anticipated (solid line) and experimental (dash dotted line) differential capacity plots (dQ/dV vs. V) during the formation step of EC and FEC electrolyte. The computed profile is obtained by convoluting the calculated reduction potentials with a 0.1 V width concave triangular wave, the experimental one is reproduced from Xia et al. [108] by removing the background.

potential as Li⁺ + coordinated EC, which, as the majority solvent, will then dominate the SEI formation process. We also note that, as Li⁺-coordinated FEC minority species are decomposing at higher potentials, the bulk equilibrium will shift to maintain the population, hence supplying the reaction.

The reduction of the Li⁺ - PF₆⁻ contact ion pair results in direct LiF formation. Upon geometry optimization, the PF₆⁻ structure is dynamically unstable, and the P-F bond spontaneously breaks to form PF₅⁻ and LiF (Fig. 7e). However, the Li⁺ - PF₆⁻ ion pair decomposition is expected to yield high reaction barrier at potential larger than 0.5 V [111]. When explicit solvent molecules are considered in the contact ion pair model, the reduction of the complex results in solvent reduction, elucidating that the decomposition of solvent molecules are preferred over that of the ion-paired PF₆⁻ (Fig. 7f, g, and n), with similar reduction potential asSSIP structures. Thus, FEC will be reduced before

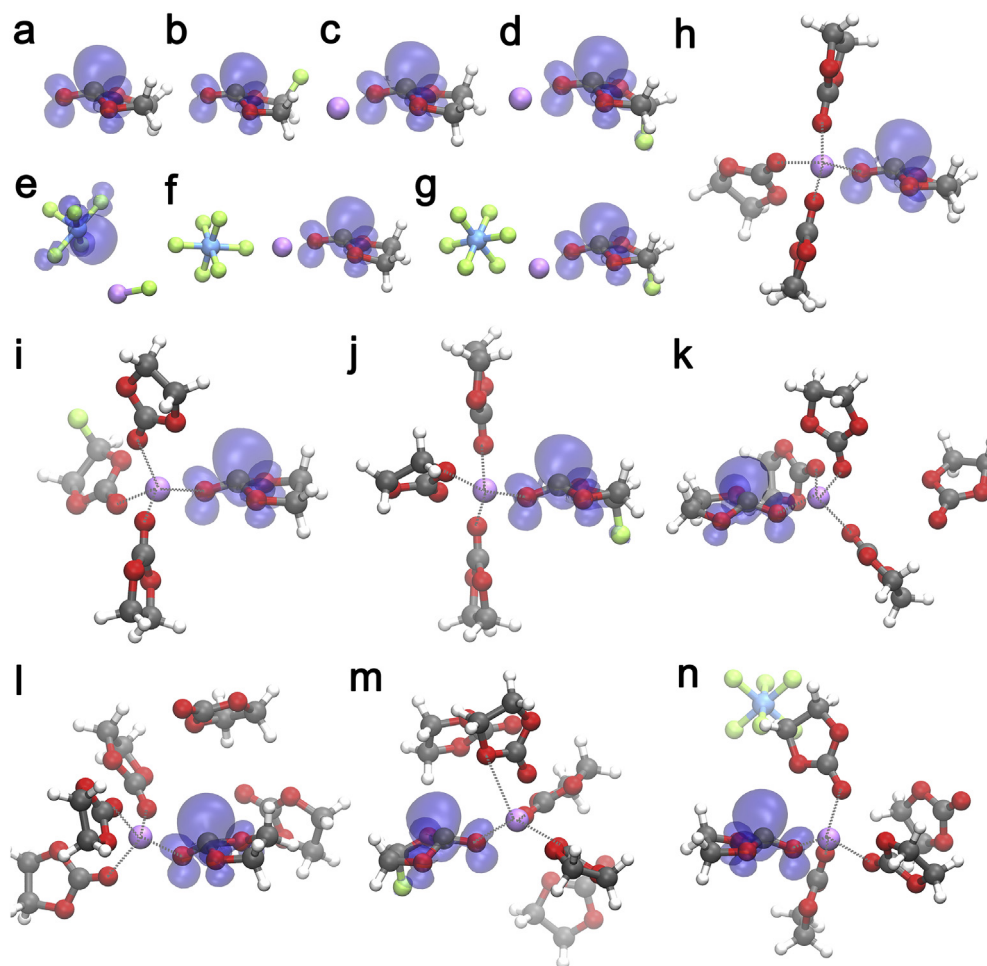
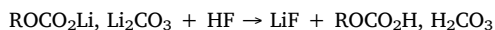
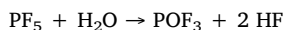
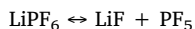


Fig. 7. Geometries and spin density analysis of reduced solvent molecules (a) EC, (b) FEC, and solvate complexes (c) $\text{Li}^+ - \text{EC}$, (d) $\text{Li}^+ - \text{FEC}$, (e) $\text{Li}^+ - \text{PF}_6^-$, (f) $\text{Li}^+ - \text{PF}_6^-(\text{EC})$, (g) $\text{Li}^+ - \text{PF}_6^-(\text{FEC})$, (h) $\text{Li}^+ - (\text{EC})_4$, (i) $\text{Li}^+ - (\text{EC})_3(\text{FEC})$ (EC reduction), (j) $\text{Li}^+ - (\text{EC})_3(\text{FEC})$ (FEC reduction), (k) $\text{Li}^+ - (\text{EC})_5$, (l) $\text{Li}^+ - (\text{EC})_6$, (m) $\text{Li}^+ - (\text{EC})_5(\text{FEC})$ (FEC reduction), and (n) $\text{Li}^+ - \text{PF}_6^-(\text{EC})_5$.

other species considered here. The reduction of PF_6^- is not expected to occur except at very high overpotentials, possibly at the final stage of SEI formation [111]. Therefore, we suggest that FEC is the major contributor to the formation of the SEI LiF as compared to PF_6^- . It is also speculated that the decomposition of PF_6^- is related to H_2O contamination rather than electrochemical reactions [52]. The degradation mechanism for LiPF_6 was suggested to be [112]:



On one hand, a higher CIP ratio in the presence of FEC would facilitate the formation of LiF. Increased amount of $\text{Li}^+ - \text{PF}_6^-$ ion pairs would promote PF_5 formation and pronounced hydrolysis, and the as-formed HF would subsequently react with the SEI carbonate species, which are initial reduction products, to form LiF [113]. Even if the trace amount of water is ignored, FEC is prone to defluorination in the presence of a Lewis acid like PF_5 , and subsequently generates F species [114]. Consequently, a greater amount of inorganic components in SEI, introduced by the sacrificial anion decomposition [115,116] or FEC defluorination, is believed to result in improved Li^+ cation transport. However, excessive HF formation under elevated temperatures may also cause SEI destruction [114], such that the operating temperature of the FEC-containing electrolyte should be controlled to avoid detrimental effect. On the other hand, an early onset of FEC reduction allows

for rapid passivation at a higher potential than EC which may limit PF_6^- reduction/decomposition. Therefore, the influence of FEC on PF_6^- decomposition is non-trivial to deconvolute. Future experimental and computational research, e.g. isotopic labelling, is recommended.

In summary, despite limited FEC concentration, we find that its presence significantly influences the properties of the electrolyte via three main mechanisms: (i) the Li^+ -coordinated FEC exhibits higher reduction potential than corresponding coordinated EC species and uncoordinated FEC, and hence contributes to an early onset of anode SEI formation and passivation, and (ii) the preferred reduction of FEC introduces a higher ratio of LiF to the SEI as compared to the EC equivalent electrolyte, and lastly (iii) a higher CIP ratio may possibly lead to increased LiF formation. The exothermic nature of the FEC reduction also supports the reported reaction pathway of defluorination and subsequent polymerization. Therefore, the origin of the excellent performance of FEC containing electrolyte may be attributed to the higher reduction potential to enable early passivation [117], substituents that promotes inorganic product formation [118], and metastable intermediates to facilitate alternative reaction pathways (such as polymerization) [27,35].

5. Conclusions

The influence of FEC on LiPF_6/EC electrolytes is investigated through classical MD simulations, FTIR experiments and first-principles calculations. The calculated solvation structure corroborates well with

the liquid structure information inferred by experiments. While select previous work advocate Li^+ coordination numbers closer to 4, both theoretical and experimental results presented here support a Li^+ solvent coordination number of 5–6 for 1.0/1.2 M LiPF_6/EC electrolytes, with or without FEC. However, reduction potentials are found to exhibit only weak dependence on the explicit number of coordinating solvents, such that 4-fold as well as 6-fold structures show similar values. Furthermore, while it is widely assumed that electrolyte additives remain largely uncoordinated in LIB electrolytes, we find that FEC, as a minority species, significantly modifies the solvation structure and reduction behavior of the electrolyte while being innocuous to the transport properties of the electrolyte. Even limited 5–10% addition of FEC results in a notably higher CIP ratio (14%, 1.0 M EC w/10% FEC) as compared to the parent EC electrolyte (6%, 1.0 M EC). FEC itself, as a fluorine-containing species, appears in the solvate complex, in 19% of the Li^+ first solvation shells (1.0 M EC w/10%FEC). We find that the Li^+ -coordinated FEC is preferentially reduced at higher reduction potentials (about 0.3 V higher than corresponding EC clusters and uncoordinated FEC), which provides early onset SEI formation and passivation of the anode surface. Meanwhile, the as-formed reduction products of FEC include a higher ratio of LiF as compared to the EC equivalent electrolyte, and a higher CIP ratio due to FEC addition may further benefit LiF formation, leading to enhanced electrochemical performance. By elucidating the solvation structure of the FEC additive in LiPF_6/EC , and its effect on the reduction potentials of the composite electrolyte, we hope to improve our understanding of the SEI-formation and its subtle dependence on the detailed intermolecular interactions and resulting solvation structure of the electrolyte.

Conflicts of interest

There are no conflicts of interest to declare.

Acknowledgements

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Appendix A. Supplementary data

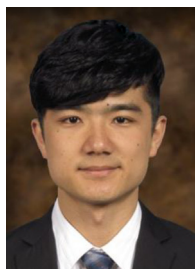
Supplementary data to this article can be found online at <https://doi.org/10.1016/j.nanoen.2019.103881>.

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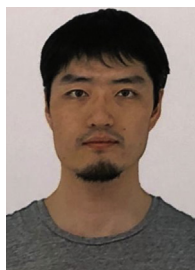
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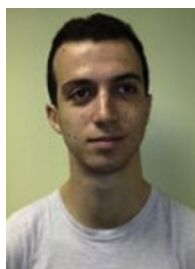
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